Coordinate Geometry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Distance and midpoint**

Find the distance between $A(1, 3)$ and $B(7, 11)$.

$AB = \sqrt{(7-1)^2 + (11-3)^2} = \sqrt{36+64} = \sqrt{100} = 10$.

2 **[Foundational]** • **Target: Distance and midpoint**

Find the midpoint of the segment joining $(-3, 5)$ and $(9, -1)$.

Midpoint $= \left(\frac{-3+9}{2}, \frac{5+(-1)}{2}\right) = (3, 2)$.

3 **[Foundational]** • **Target: Distance and midpoint**

The midpoint of $A(2, -3)$ and $B(x, y)$ is $(5, 1)$. Find the coordinates of B.

$\frac{2+x}{2} = 5$ gives $x = 8$. $\frac{-3+y}{2} = 1$ gives $y = 5$. So $B = (8, 5)$.

4 **[Foundational]** • **Target: Gradient and equations of lines**

Find the gradient of the line through $(2, -1)$ and $(6, 7)$.

Gradient $= \frac{7-(-1)}{6-2} = \frac{8}{4} = 2$.

5 **[Foundational]** • **Target: Gradient and equations of lines**

Find the equation of the line through $(4, 3)$ with gradient -2, in the form $y = mx + c$.

$y - 3 = -2(x - 4)$, so $y = -2x + 8 + 3 = -2x + 11$.

6 **[Foundational]** • **Target: Gradient and equations of lines**

Find the equation of the line through $(-1, 4)$ and $(3, -4)$, in the form $y = mx + c$.

Gradient $= \frac{-4-4}{3-(-1)} = \frac{-8}{4} = -2$. $y - 4 = -2(x + 1)$, so $y = -2x + 2$.

7 **[Foundational]** • **Target: Parallel and perpendicular lines**

Find the equation of the line parallel to $y = 3x - 1$ passing through $(5, 0)$.

Parallel lines share the same gradient, $m = 3$: $y - 0 = 3(x - 5)$, so $y = 3x - 15$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Parallel and perpendicular lines

Find the equation of the line perpendicular to $y = \frac{1}{3}x + 2$ passing through $(6, -1)$.

The perpendicular gradient is the negative reciprocal: $m = -3$. $y - (-1) = -3(x - 6)$, so $y = -3x + 18 - 1 = -3x + 17$.

9 [Intermediate] • Target: Parallel and perpendicular lines

Determine whether the lines $y = 4x - 3$ and $4y + x = 8$ are parallel, perpendicular, or neither.

Rewrite the second line as $y = -\frac{1}{4}x + 2$, gradient $-\frac{1}{4}$. The product of the gradients is $4 \times \left(-\frac{1}{4}\right) = -1$, so the lines are perpendicular.

10 [Intermediate] • Target: The equation of a circle

Write the equation of a circle with center $(3, -2)$ and radius 6.

$(x - 3)^2 + (y + 2)^2 = 36$.

11 [Intermediate] • Target: The equation of a circle

Find the center and radius of $x^2 + y^2 - 4x + 6y - 12 = 0$ by completing the square.

Group: $(x^2 - 4x) + (y^2 + 6y) = 12$. Complete the square: $(x - 2)^2 - 4 + (y + 3)^2 - 9 = 12$, so $(x - 2)^2 + (y + 3)^2 = 25$. Center: $(2, -3)$. Radius: $\sqrt{25} = 5$.

12 [Intermediate] • Target: The equation of a circle

A circle has equation $(x + 1)^2 + (y - 4)^2 = 49$. State its center and radius.

Center: $(-1, 4)$. Radius: $\sqrt{49} = 7$.

13 [Intermediate] • Target: Lines and circles together

A circle has center $(0, 0)$ and radius 5. Determine whether the line $y = x + 8$ intersects the circle, is tangent to it, or misses it entirely, by comparing the distance from the center to the line against the radius.

Rewrite the line as $x - y + 8 = 0$. Distance from $(0, 0)$: $d = \frac{|0 - 0 + 8|}{\sqrt{1^2 + (-1)^2}} = \frac{8}{\sqrt{2}} = 4\sqrt{2} \approx 5.66$. Since $5.66 > 5$ (the distance exceeds the radius), the line misses the circle entirely.

14 [Intermediate] • Target: Lines and circles together

A circle has center $(2, 1)$ and radius 3. Determine whether the line $x + y = 10$ intersects the circle, is tangent to it, or misses it entirely.

Rewrite the line as $x + y - 10 = 0$. Distance from $(2, 1)$: $d = \frac{|2 + 1 - 10|}{\sqrt{1^2 + 1^2}} = \frac{7}{\sqrt{2}} = \frac{7\sqrt{2}}{2} \approx 4.95$. Since $4.95 > 3$, the line misses the circle entirely.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: The perpendicular bisector** • **Cross-Lesson: 18.01 Binomial Distribution**

Find the equation of the perpendicular bisector of the segment joining $A(1, 2)$ and $B(7, 8)$.

Midpoint: $(4, 5)$. Gradient of AB: $\frac{8-2}{7-1} = 1$. Perpendicular gradient: -1. Equation: $y - 5 = -1(x - 4)$, so $y = -x + 9$.

16 **[Challenging]** • **Target: The perpendicular bisector** • **Cross-Lesson: 19.01 Normal Distribution**

Find the equation of the perpendicular bisector of the segment joining $A(-2, 3)$ and $B(4, -5)$.

Midpoint: $(1, -1)$. Gradient of AB: $\frac{-5-3}{4-(-2)} = \frac{-8}{6} = -\frac{4}{3}$. Perpendicular gradient: $\frac{3}{4}$. Equation: $y + 1 = \frac{3}{4}(x - 1)$, so $y = \frac{3}{4}x - \frac{7}{4}$.

17 **[Challenging]** • **Target: Area of a triangle from coordinates** • **Cross-Lesson: 2.01 Simultaneous Equations & Inequalities**

Find the area of the triangle with vertices $P(0, 0)$, $Q(8, 0)$, $R(3, 6)$.

Using the coordinate area formula: Area $= \frac{1}{2}|x_P(y_Q - y_R) + x_Q(y_R - y_P) + x_R(y_P - y_Q)| = \frac{1}{2}|0(0-6) + 8(6-0) + 3(0-0)| = \frac{1}{2}(48) = 24$.

18 **[Challenging]** • **Target: Area of a triangle from coordinates** • **Cross-Lesson: 3.01 Graphs & Transformations**

Find the area of the triangle with vertices $A(-3, 2)$, $B(5, 2)$, $C(1, 7)$.

Area $= \frac{1}{2}|x_A(y_B - y_C) + x_B(y_C - y_A) + x_C(y_A - y_B)| = \frac{1}{2}|(-3)(2-7) + 5(7-2) + 1(2-2)| = \frac{1}{2}|15 + 25 + 0| = \frac{1}{2}(40) = 20$.

19 **[Challenging]** • **Target: A well-designed word problem: locating a cell tower** • **Cross-Lesson: 5.01 Trigonometry**

This question has two parts. Two houses are located at $A(0, 0)$ and $B(12, 0)$ (positions in kilometers). A cell tower must be built equidistant from both houses, and it must also lie on the line $y = x + 2$. (a) Find the equation of the perpendicular bisector of AB, and use it together with $y = x + 2$ to find the tower's coordinates. (b) Find the distance from the tower to each house, and confirm they are equal.

(a) Midpoint of AB: $(6, 0)$. Since AB is horizontal, its perpendicular bisector is the vertical line $x = 6$. Substituting into $y = x + 2$: $y = 8$. Tower location: $(6, 8)$. (b) Distance to A: $\sqrt{(6-0)^2 + (8-0)^2} = \sqrt{36 + 64} = \sqrt{100} = 10$ km. Distance to B: $\sqrt{(6-12)^2 + (8-0)^2} = \sqrt{36 + 64} = \sqrt{100} = 10$ km. Both distances equal 10 km, confirming the tower is equidistant from both houses.

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20 [Challenging] • Target: A well-designed word problem: locating a cell tower • Cross-Lesson: 6.01 Differentiation & Integration

The diagram shows a triangle ABC with coordinates A(2, 1), B(8, 1) and C(5, 5). Find the length of the side AC.

Final Answer: x. The coordinates of A are (2, 1) and the coordinates of C are (5, 5). Using the distance formula, $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. Length of AC = $\sqrt{(5 - 2)^2 + (5 - 1)^2}$ Length of AC = $\sqrt{(3)^2 + (4)^2}$ Length of AC = $\sqrt{9 + 16}$ Length of AC = $\sqrt{25}$ Length of AC = 5.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Lines, Lengths & Midpoints

Find the equation of the line passing through $A(2, -3)$ and $B(6, 5)$ in the form $y = mx + c$.

a. $y = 2x - 7$ ✓

b. $y = 2x + 1$

c. $y = \frac{1}{2}x - 4$

d. $y = 2x - 3$

Slope $m = \frac{5 - (-3)}{6 - 2} = \frac{8}{4} = 2$. Line: $y - 5 = 2(x - 6) \Rightarrow y = 2x - 7$.

22 [Foundational] • Target: Lines, Lengths & Midpoints

Find the length of the line segment joining $P(-1, 4)$ and $Q(5, -4)$.

a. **10** ✓

b. 8

c. $2\sqrt{7}$

d. 12

$d = \sqrt{(5 - (-1))^2 + (-4 - 4)^2} = \sqrt{6^2 + (-8)^2} = \sqrt{36 + 64} = 10$.

23 [Foundational] • Target: Lines, Lengths & Midpoints

Find the midpoint of the segment connecting $C(-3, 8)$ and $D(7, -2)$.

a. $(2, 3)$ ✓

b. $(5, 3)$

c. $(2, 5)$

d. $(4, 6)$

$M = \left(\frac{-3 + 7}{2}, \frac{8 + (-2)}{2}\right) = (2, 3)$.

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24 [Foundational] • Target: Perpendicular Bisectors & Gradients

Find the equation of the perpendicular bisector of the segment joining $A(1, 2)$ and $B(5, 10)$.

a. $x + 2y = 15$ ✓

b. $2x - y = 0$

c. $x - 2y = -9$

d. $2x + y = 12$

Midpoint $M = (3, 6)$. Gradient of $AB = \frac{10-2}{5-1} = 2$. Perpendicular gradient $m_{\perp} = -1/2$. Equation: $y - 6 = -\frac{1}{2}(x - 3) \Rightarrow 2y - 12 = -x + 3 \Rightarrow x + 2y = 15$.

25 [Foundational] • Target: Perpendicular Bisectors & Gradients

The line L_1 has equation $3x - 4y = 12$. Find the equation of the line L_2 perpendicular to L_1 passing through $(2, 1)$.

a. $4x + 3y = 11$ ✓

b. $3x + 4y = 10$

c. $4x - 3y = 5$

d. $3x - 4y = 2$

Slope of L_1 is $3/4$. Perpendicular slope is $-4/3$. Equation: $y - 1 = -\frac{4}{3}(x - 2) \Rightarrow 3y - 3 = -4x + 8 \Rightarrow 4x + 3y = 11$.

26 [Foundational] • Target: Perpendicular Bisectors & Gradients

Three points are given: $P(1, 1)$, $Q(4, 5)$, and $R(k, 9)$. If P, Q, R are collinear, find k.

a. **7** ✓

b. 6

c. 8

d. 5

Gradient $PQ = \frac{5-1}{4-1} = \frac{4}{3}$. Gradient $QR = \frac{9-5}{k-4} = \frac{4}{k-4} = \frac{4}{3} \Rightarrow k - 4 = 3 \Rightarrow k = 7$.

27 [Foundational] • Target: Circles — Equations, Tangents & Normals

Find the center and radius of the circle given by $x^2 + y^2 - 6x + 8y - 11 = 0$.

a. **Center** $(3, -4)$**, radius 6** ✓

b. Center $(-3, 4)$, radius 6

c. Center $(3, -4)$, radius $\sqrt{11}$

d. Center $(-3, 4)$, radius 36

$(x-3)^2 - 9 + (y+4)^2 - 16 - 11 = 0 \Rightarrow (x-3)^2 + (y+4)^2 = 36$. Center is $(3, -4)$, radius is $\sqrt{36} = 6$.

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28 [Foundational] • Target: Circles — Equations, Tangents & Normals

Find the equation of the tangent to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$.

a. $3x + 4y = 25$ ✓

b. $4x - 3y = 0$

c. $3x - 4y = 25$

d. $4x + 3y = 24$

The radius to $(3, 4)$ has gradient $4/3$. Tangent is perpendicular, with gradient $-3/4$. Equation: $y - 4 = -\frac{3}{4}(x - 3) \Rightarrow 4y - 16 = -3x + 9 \Rightarrow 3x + 4y = 25$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Circles — Equations, Tangents & Normals

A circle has diameter with endpoints $A(-2, 1)$ and $B(4, 9)$. Find the equation of the circle.

a. $(x - 1)^2 + (y - 5)^2 = 25$ ✓

b. $(x + 1)^2 + (y + 5)^2 = 25$

c. $(x - 1)^2 + (y - 5)^2 = 100$

d. $(x - 3)^2 + (y - 4)^2 = 25$

Center $M = \left(\frac{-2+4}{2}, \frac{1+9}{2}\right) = (1, 5)$. Radius $r = \frac{1}{2}\sqrt{(4 - (-2))^2 + (9 - 1)^2} = \frac{1}{2}\sqrt{36 + 64} = 5$. Equation: $(x - 1)^2 + (y - 5)^2 = 25$.

30 [Intermediate] • Target: Intersections & Exam-Standard Geometry

Determine the relationship between the line $y = x - 5$ and the circle $x^2 + y^2 = 9$.

a. **The line does not intersect the circle** ✓

b. The line is tangent to the circle

c. The line intersects the circle in two distinct points

d. The line passes through the center of the circle

Substitute: $x^2 + (x - 5)^2 = 9 \Rightarrow 2x^2 - 10x + 16 = 0 \Rightarrow x^2 - 5x + 8 = 0$. $\Delta = 25 - 32 = -7 < 0$. No real solutions, so no intersection.

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31 [Intermediate] • Target: Intersections & Exam-Standard Geometry

Find the area of the triangle with vertices $A(0,0)$, $B(6,2)$, and $C(2,8)$.

a. **22** ✓

b. 20

c. 24

d. 18

Area $= \frac{1}{2}|x_A(y_B - y_C) + x_B(y_C - y_A) + x_C(y_A - y_B)| = \frac{1}{2}|0 + 6(8-0) + 2(0-2)| = \frac{1}{2}|48-4| = 22$.

32 [Intermediate] • Target: Intersections & Exam-Standard Geometry

Find the length of the chord cut off from the line $y = 2$ by the circle $x^2 + y^2 = 13$.

a. **6** ✓

b. 3

c. $2\sqrt{13}$

d. 8

Set $y = 2$: $x^2 + 4 = 13 \Rightarrow x^2 = 9 \Rightarrow x = \pm 3$. The points are $(-3, 2)$ and $(3, 2)$, so chord length is $3 - (-3) = 6$.

33 [Intermediate] • Target: Distance and midpoint

A circle has equation $(x-7)^2 + (y+2)^2 = 169$. State its centre and radius.

a. **centre** $(7, -2)$**, radius** 13 ✓

b. centre $(-2, 7)$, radius 13

c. centre $(-7, 2)$, radius 13

d. centre $(7, -2)$, radius 169

Comparing to $(x-a)^2 + (y-b)^2 = r^2$: centre $(7, -2)$, and $r = \sqrt{169} = 13$.

34 [Intermediate] • Target: Gradient and equations of lines

A circle has equation $(x+5)^2 + (y-3)^2 = 100$. State its centre and radius.

a. **centre** $(-5, 3)$**, radius** 10 ✓

b. centre $(-5, 3)$, radius 100

c. centre $(3, -5)$, radius 10

d. centre $(5, -3)$, radius 10

Comparing to $(x-a)^2 + (y-b)^2 = r^2$: centre $(-5, 3)$, and $r = \sqrt{100} = 10$.

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35 [Intermediate] • Target: Parallel and perpendicular lines

A circle has equation $(x-3)^2+(y+3)^2=100$. State its centre and radius.

a. **centre** $(3,-3)$**, radius** 10 ✓

b. centre $(4,-3)$, radius 10

c. centre $(3,-3)$, radius 100

d. centre $(-3,3)$, radius 10

Comparing to $(x-a)^2+(y-b)^2=r^2$: centre $(3,-3)$, and $r=\sqrt{100}=10$.

36 [Intermediate] • Target: The equation of a circle

A circle has equation $(x-3)^2+(y+5)^2=100$. State its centre and radius.

a. centre $(-3,5)$, radius 10

b. **centre** $(3,-5)$**, radius** 10 ✓

c. centre $(-5,3)$, radius 10

d. centre $(3,-5)$, radius 100

Comparing to $(x-a)^2+(y-b)^2=r^2$: centre $(3,-5)$, and $r=\sqrt{100}=10$.

37 [Intermediate] • Target: Lines and circles together

A circle has equation $(x-5)^2+(y+5)^2=81$. State its centre and radius.

a. centre $(5,-5)$, radius 81

b. **centre** $(5,-5)$**, radius** 9 ✓

c. centre $(-5,5)$, radius 9

d. centre $(6,-5)$, radius 9

Comparing to $(x-a)^2+(y-b)^2=r^2$: centre $(5,-5)$, and $r=\sqrt{81}=9$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The perpendicular bisector • Cross-Lesson: 3.01 Graphs & Transformations

A circle has equation $(x+1)^2+(y-7)^2=25$. State its centre and radius.

a. centre $(7,-1)$, radius 5

b. centre $(-1,7)$, radius 25

c. centre $(1,-7)$, radius 5

d. **centre** $(-1,7)$**, radius** 5 ✓

Comparing to $(x-a)^2+(y-b)^2=r^2$: centre $(-1,7)$, and $r=\sqrt{25}=5$.

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39 [Challenging] • Target: Area of a triangle from coordinates • Cross-Lesson: 5.01 Trigonometry

A circle has equation $(x-7)^2+(y-5)^2=25$. State its centre and radius.

a. centre $(5, 7)$, radius 5

b. **centre** $(7, 5)$**, radius** 5 ✓

c. centre $(-7, -5)$, radius 5

d. centre $(7, 5)$, radius 25

Comparing to $(x-a)^2+(y-b)^2=r^2$: centre $(7, 5)$, and $r=\sqrt{25}=5$.

40 [Challenging] • Target: A well-designed word problem: locating a cell tower • Cross-Lesson: 6.01 Differentiation & Integration

A circle has equation $(x-7)^2+(y-8)^2=16$. State its centre and radius.

a. centre $(7, 8)$, radius 16

b. centre $(-7, -8)$, radius 4

c. **centre** $(7, 8)$**, radius** 4 ✓

d. centre $(8, 7)$, radius 4

Comparing to $(x-a)^2+(y-b)^2=r^2$: centre $(7, 8)$, and $r=\sqrt{16}=4$.

41 [Challenging] • Target: Lines, Lengths & Midpoints • Cross-Lesson: 7.01 Statistics & Probability

A circle has equation $(x-6)^2+(y-5)^2=16$. State its centre and radius.

a. centre $(6, 5)$, radius 16

b. centre $(-6, -5)$, radius 4

c. centre $(5, 6)$, radius 4

d. **centre** $(6, 5)$**, radius** 4 ✓

Comparing to $(x-a)^2+(y-b)^2=r^2$: centre $(6, 5)$, and $r=\sqrt{16}=4$.

42 [Challenging] • Target: Perpendicular Bisectors & Gradients • Cross-Lesson: 8.01 Functions: Composite & Inverse

Find the equation of the circle with centre $(-7, -2)$ and radius 12.

a. $(x+7)^2+(y+2)^2=144$ ✓

b. $(x-7)^2+(y-2)^2=144$

c. $(x+7)^2+(y+2)^2=12$

d. $(x+7)^2+(y+2)^2=148$

Circle formula $(x-a)^2+(y-b)^2=r^2$ with $(a, b)=(-7, -2)$, $r=12$: $(x+7)^2+(y+2)^2=144$.

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43 [Challenging] • Target: Circles — Equations, Tangents & Normals • Cross-Lesson: 9.01 Circular Measure

Find the equation of the circle with centre $(-5, 2)$ and radius 11.

a. $(x+5)^2+(y-2)^2=125$

b. $(x+5)^2+(y-2)^2=11$

c. $(x-5)^2+(y+2)^2=121$

d. $(x+5)^2+(y-2)^2=121$ ✓

Circle formula $(x-a)^2+(y-b)^2=r^2$ with $(a,b)=(-5,2)$, $r=11$: $(x+5)^2+(y-2)^2=121$.

44 [Challenging] • Target: Intersections & Exam-Standard Geometry • Cross-Lesson: 1.01 Indices, Surds & Quadratics

Find the equation of the circle with centre $(-3,\ -3)$ and radius 4.

a. $(x+3)^2+(y+3)^2=20$

b. $(x+3)^2+(y+3)^2=4$

c. $(x+3)^2+(y+3)^2=16$ ✓

d. $(x-3)^2+(y-3)^2=16$

Circle formula $(x-a)^2+(y-b)^2=r^2$ with $(a,b)=(-3,\ -3)$, $r=4$: $(x+3)^2+(y+3)^2=16$.

45 [Challenging] • Target: Distance and midpoint • Cross-Lesson: 10.01 Series & Binomial Theorem

Find the equation of the circle with centre $(-2,\ -6)$ and radius 6.

a. $(x+2)^2+(y+6)^2=36$ ✓

b. $(x+2)^2+(y+6)^2=40$

c. $(x+2)^2+(y+6)^2=6$

d. $(x-2)^2+(y-6)^2=36$

Circle formula $(x-a)^2+(y-b)^2=r^2$ with $(a,b)=(-2,\ -6)$, $r=6$: $(x+2)^2+(y+6)^2=36$.

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